

Problem 1. Let $W_t \in \mathbb{R}^{IJ}$. Suppose we are given an initial vector x_0 . Define $x_{t+1} = W_t x_t$.

part a:. Rewrite equation (1) in terms of the tensors using summation notation.

Proof.

$$x[t, j] = \sum_i W[t, i, j] x[t-1, i]$$

□

part b:. Rewrite equation (1) in terms of the tensors using Einstein summation notation.

Proof.

$$x[t, J] = W[t, I, J] x[t-1, I]$$

□

Problem 2. Consider convolving a kernel $K[n_{out}, \Delta x, \Delta y, n_{in}]$ won a layer $L[b, x, y, n_{in}]$, where $B, X, Y, N_{in}, N_{out}, \Delta x, \Delta y$ are number of possible values. How many floating point multiples are required to compute the convolution on the batch?

Proof.

$$BXY\Delta X\Delta YN_{in}N_{out}$$

□

Problem 3. Suppose that we want a video CNN producing layers of the form $L[b, x, y, t, n]$. Write the equation for computing $L_{\ell+1}[b, x, y, t, n]$ from the tensor $L_\ell[B, X, Y, T, I]$. The filter should include an index Δt and handle a stride s applied to both space and time.

Proof.

$$\begin{aligned} L_{\ell+1}[b, x, y, t, j] \\ = \sigma(K_{\ell+1}[j, \Delta X, \Delta Y, \Delta T, I] L_\ell[b, s \cdot x + \Delta X, s \cdot y + \Delta Y, s \cdot t + \Delta T, I] - B[j]) \end{aligned}$$

□

Problem 5. Scale Invariance. Consider a batch of input images

$$L_{0,d}[b, x, y, n]$$

$d = 2^k$ is the spatial dimension of x and y and n ranges over the three color channels. To capture scale invariance we compute a set of layers $L_{\ell,d}$ with $0 \leq \ell \leq \ell_{max}$ and d a power of two with $4 \leq d \leq d_{max}$ where d_{max} is the spatial dimension of the input images. We set $d_{min} = 4$ to allow 3×3 convolutional filters to the lowest spatial resolution. The output layer, say for image classification, is $L_{\ell_{max}, d_{min}}[b, x, y, n]$

We define $L_{0,d}[b, x, y, n]$ to be a layer in an "image pyramid" constructed by successively down-sampling the images by a factor of two.

$$L_{0,d/2}[b, x, y, n] = \mathbb{E}_{x', y' \sim \text{Ber}(1/2)} L_{0,d}[b, 2x + x', 2y + y', n]$$

We then define $L_{\ell+1, d_{max}}[b, x, y, n]$ by 3×3 convolutions that do not change the image dimension, i.e.

$$L_{\ell+1, d_{max}}[b, x, y, n] = \sigma(K_{\ell+1}[n_{out}, \Delta X, \Delta Y, N_{in}] L_{\ell, d_{max}}[b, x + \Delta X, y + \Delta Y, N_{in}] - B_{\ell+1}[n_{out}])$$

For $d < d_{max}$ give an equation for computing $L_{\ell+1, d}[b, x, y, n]$ from $L_{\ell, d}[b, x, y, n]$ and $L_{\ell, 2d}[b, x, y, n]$ using the same kernel $K_{\ell+1}[n_{out}, \Delta X, \Delta Y, N_{in}]$ for both inputs.

Proof.

$$\begin{aligned} L_{\ell+1, d}[b, x, y, n] &= \sigma(K_{\ell+1}[n, \Delta X, \Delta Y, N_{in}] L_{\ell, d}[b, x + \Delta X, y + \Delta Y, N_{in}] \\ &\quad + K_{\ell+1}[n, \Delta X, \Delta Y, N_{in}] L_{\ell, 2d}[b, 2x + \Delta X, 2y + \Delta Y, N_{in}] - B_{\ell+1}[n]) \end{aligned}$$

□